ON A NOETHERIAN ALGEBRAIC STACK EVERY QC MODULE IS A FILTERED COLIMIT OF COHERENT SUBMODULES

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Coherent submodules of qc modules. Cohomology of Algebraic Stacks, Lemma stacks-cohomology-lemma-coherent-Noetherian-quasi-coherent-subquotient. Images and sums of coherent submodules are coherent.
- Groupoid presentation of QCoh and Coh. Cohomology of Algebraic Stacks, Lemma stacks-cohomology-lemma-coherent-presentation; Algebraic Stacks, Lemma algebraic-lemma-stack-presentation. Identifies the categories with groupoid modules.
- Filtered colimits of coherent modules on groupoids. Groupoids in Spaces, Lemma spaces-groupoids-lemma-colimit-coherent. The groupoid-level statement being transported.

Source reference. Original lemma 0GRB: sources/stacks-chow/tags.txt, source lines 20891–20891. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04YC: sources/stacks-chow/tags.txt, source lines 6425–6425. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04T5: sources/stacks-chow/tags.txt, source lines 6243–6243. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0GRC: sources/stacks-chow/tags.txt, source lines 20892–20892. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0GPS: sources/stacks-chow/tags.txt, source lines 20837–20837. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04YG: sources/stacks-chow/tags.txt, source lines 6429–6429. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04YH: sources/stacks-chow/tags.txt, source lines 6430–6430. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:stacks-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 075S: sources/stacks-chow/tags.txt, source lines 9141–9141. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04ZL: sources/stacks-chow/tags.txt, source lines 6469–6469. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\mathcal{X}$ be a Noetherian algebraic stack. Every quasi-coherent $\mathcal{O}_\mathcal{X}$-module is the filtered colimit of its coherent submodules.*

Proof. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_\mathcal{X}$-module. If $\mathcal{G}, \mathcal{H} \subset \mathcal{F}$ are coherent $\mathcal{O}_\mathcal{X}$-submodules then the image of $\mathcal{G} \oplus \mathcal{H} \to \mathcal{F}$ is another coherent $\mathcal{O}_\mathcal{X}$-submodule which contains both of them, see Lemma 0GRB. In this way we see that the system is directed. Hence it now suffices to show that $\mathcal{F}$ can be written as a filtered colimit of coherent modules, as then we can take the images of these modules in $\mathcal{F}$ to conclude there are enough of them.

Let U be an affine scheme and $U \to \mathcal{X}$ a surjective smooth morphism (Properties of Stacks, Lemma 04YC). Set $R = U \times_\mathcal{X} U$ so that $\mathcal{X} = [U/R]$ as in Algebraic Stacks, Lemma 04T5. By Lemma 0GRC we have $QCoh(\mathcal{O}_X) = QCoh(U, R, s, t, c)$ and $Coh(\mathcal{O}_X) = Coh(U, R, s, t, c)$. In this way we reduce to the problem of proving the corresponding thing for $QCoh(U, R, s, t, c)$. This is Groupoids in Spaces, Lemma 0GPS; we check its assumptions in the next paragraph.

We urge the reader to skip the rest of the proof. The affine scheme U is Noetherian; this follows from our definition of $\mathcal{X}$ being locally Noetherian, see Properties of Stacks, Definition 04YG and Remark 04YH. The projection morphisms $s, t : R \to U$ are smooth (see reference given above) and quasi-separated and quasi-compact (Morphisms of Stacks, Lemma 075S). In particular, R is a quasi-compact and quasi-separated algebraic space smooth over U and hence Noetherian (Morphisms of Spaces, Lemma 04ZL). □